

Integration of the Mobile Robot and Internet of Things to Collect Data from the Agricultural Fields

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Abstract— Robotics and Internet of Things (IoT) are two hot topics in the research area. There are studies in the literature that combine these two topics. In this study, robotics and IoT are used for the agricultural fields. Because, with the technology becoming more available on agriculture; food security, crop yield will be increased and the environmental hazards will be decreased. But this can be achieved by strictly monitoring the agricultural fields and greenhouses. For these purposes, static sensors or sensor networks, and mobile agents are used. IoT forms the backbone of such systems because there are too many units in different places and a lot of data is coming out from these units. Also, processing this data reveals the big data problem. Purpose of this work is integrating the mobile robot platform to gather data from the agricultural fields or greenhouses and then post this gathered data to the web application. So that, data can be stored, processed, and classified on the web application or cloud. This study proposes a design scheme for the mobile internet of things concept where the client or the agent is the mobile robot whether it is autonomous or not. Furthermore, the design structure is not limited to the mobile ground vehicle. Any type of unmanned vehicle or static sensor can be integrated into the system. Internet connection is not limited to only Wi-Fi, there is also a cellular connection in the system. With this study, mobile data acquisition and transferring this acquired data to the web application can be done. Also, the infrastructure of the mobile agent-based internet of things system is shown. On the robot side, autonomy can be added to the system.

Keywords—robotics, internet of things, cloud, data acquisition, precision agriculture, smart farming

I. INTRODUCTION

Precision agriculture was one of the most trending topics when the cross section of technology and agriculture come to mind. With the increase rate of smart devices, in the next decade smart agriculture might be the most trending word agriculture. With the usage of smart farming, agricultural fields can be more productive [1]. Smart farming is the intersection of internet of things and precision agriculture. This study shows how robotics can be added to this smart farming equation. Although the robotics is the main focus, static sensors or sensor networks can be operated using the proposed system.

Precision agriculture is usage of information technologies to make farms more efficient[2]. Precision agriculture can help the sustainability of agricultural production [3]. Engineering technologies have high impact on development of precision agriculture[4].Internet of things is well known concept for connecting the devices, sensors, or systems through the internet connection [5]–[8]. Internet of things has usage in agricultural fields [9], [10]. Backbone of the precision agriculture is internet

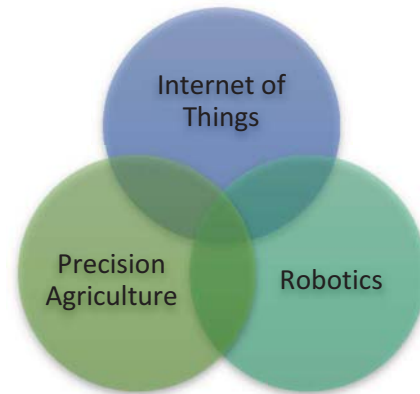


Fig. 1. Motivation of this study.

of things concept. To collect data from the agricultural fields internet of things is used. But mostly, static wireless sensor networks are used for this objective. There are some studies for agricultural robotics [11].

Another important subject is connected robots. The naming convention for robots is unmanned mobile agents. Connected robots can work in parallel. The applications of these systems are already used in industrial factories, warehouses, and defense industry. The intersection of internet of things and robotics is named as internet of robotic things [12]. With the connected robot systems, fields can be monitored from above using drones, decision making systems on cloud servers with deep learning can give tasks to mobile robots, and mobile robots with weeding, sowing, or harvesting capabilities can control the field in completely autonomous way.

This study is the continuation of the AGROBOT system [13]. Robot at the study is redesigned to be more reliable, while preserving the design requirements. The design requirements were modularity, low cost production, domestic production, suitability of environmental conditions, and originality. Future work of the study was integrating the robot with management system using internet and completing the software structure [13]. To accomplish these tasks this work has been done.

Aim of this article is giving brief but enough information to the researchers, scientist, and developers about the interdisciplinary study that covers the fields, that are shown at the Fig. 1. First the system design is explained in conceptual level. Then, the elements in the system, that are mobile robot and server, are explained in detail with schematics in the following sections. Lastly, the integration of the whole system is given.

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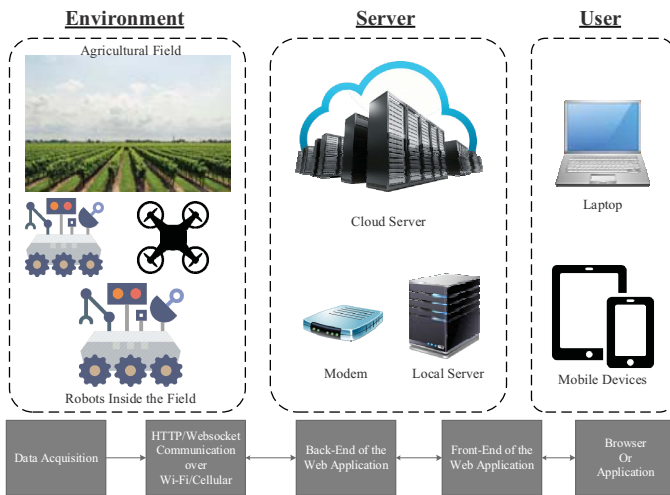


Fig. 2. Conceptual design of the system.

II. SYSTEM DESIGN

At the Fig. 2, conceptual design of the system is shown. System has two main components. First one is the mobile robot and the second one is server. Mobile robot works as a mobile agent to collect data from the field using on-board sensors. These sensors can be MEMS pressure sensor, RGB image, or spectrometer. Other type of mobile agents, such as drones can be integrated to the server. Server collects data from the mobile robot using cellular communication or Wi-Fi. In case of Wi-Fi communication, there are two configurations. In first configuration, robot directly communicates with server, established on the cloud, using cellular communication. Second configuration robot either communicates with server in local area network using Wi-Fi communication or communicates with server on the cloud through Wi-Fi modem with internet connection. Laptop and mobile represent the user interface.

III. MOBILE ROBOT ARCHITECTURE

A. Hardware

Mobile robot is the renewed version of the previous design "AGROBOT". While the old robot mechanics are same, whole electronic design is renewed. New onboard computer, stereo camera, pan-tilt camera, and cellular connection module is added. Further detail will be the subject of the new paper. In this study, sensor network and communication parts, that are on the robot, is discussed.

Robot is shown at the Fig. 3. There are six parts on the robot that are related with this system. Data acquisition system on the robot consists of Wi-Fi communication, cellular communication module, environment sensors board, GPS board, robot sensors, stereo camera, and auxiliary camera.

- Wi-Fi communication is used when the Wi-Fi network is available. This can be Wi-Fi router with internet connection. Also, server can be used on the local area network (LAN). For example, greenhouse monitoring system without internet connection. Wi-Fi communication is controlled by NVidia Jetson TX1 board.



Fig. 3. Mobile robot.

- Another internet connection is cellular module that enables the internet connection using cellular network. By using this module robot can connect to server from anywhere with cellular connection.
- Environment sensor board, consists of micro controller unit (MCU), environment sensors, and global positioning system (GPS) module. These environment sensors are low cost micro electro mechanical system (MEMS) based sensors, that are publicly available in the market. These sensors are easy to use, low cost, and small. Disadvantages of these sensors are calibration and sensitivity. They are not for the very precise measurements. In the system, these sensor measurements are directly related with agricultural field. GPS module is used for the robot navigation. Position of the measurements is very important to know where the data is collected. Besides the position, time is another important factor, that GPS module provides. GPS module gives the position with 5m resolution. Since the GPS module uses the global time system, the time relativity problems has ceased to exist. MCU board is Arduino UNO R3. MCU board is used to control the MEMS sensors and GPS module, and integrate these sensors with the rest of the robot by sending the collected data periodically.
- Robot sensors are installed on the robot. These sensors or boards measures the battery voltage, current draw, internal battery and motor driver temperatures, and self-generated navigational data, such as odometer, inertial measurement unit. Reason that, these sensors are integrated with system is keeping track of robot status.
- Stereo camera is used for the navigational purposes. It can capture stereo 2k image and generate depth map. Image is one of the most information-bearing sensor. In addition to this robot can map the field by fusing depth and generate RGB-D map.
- Auxiliary camera is placed to capture image from the right side of the field. This camera can be used to capture image from the field rows.

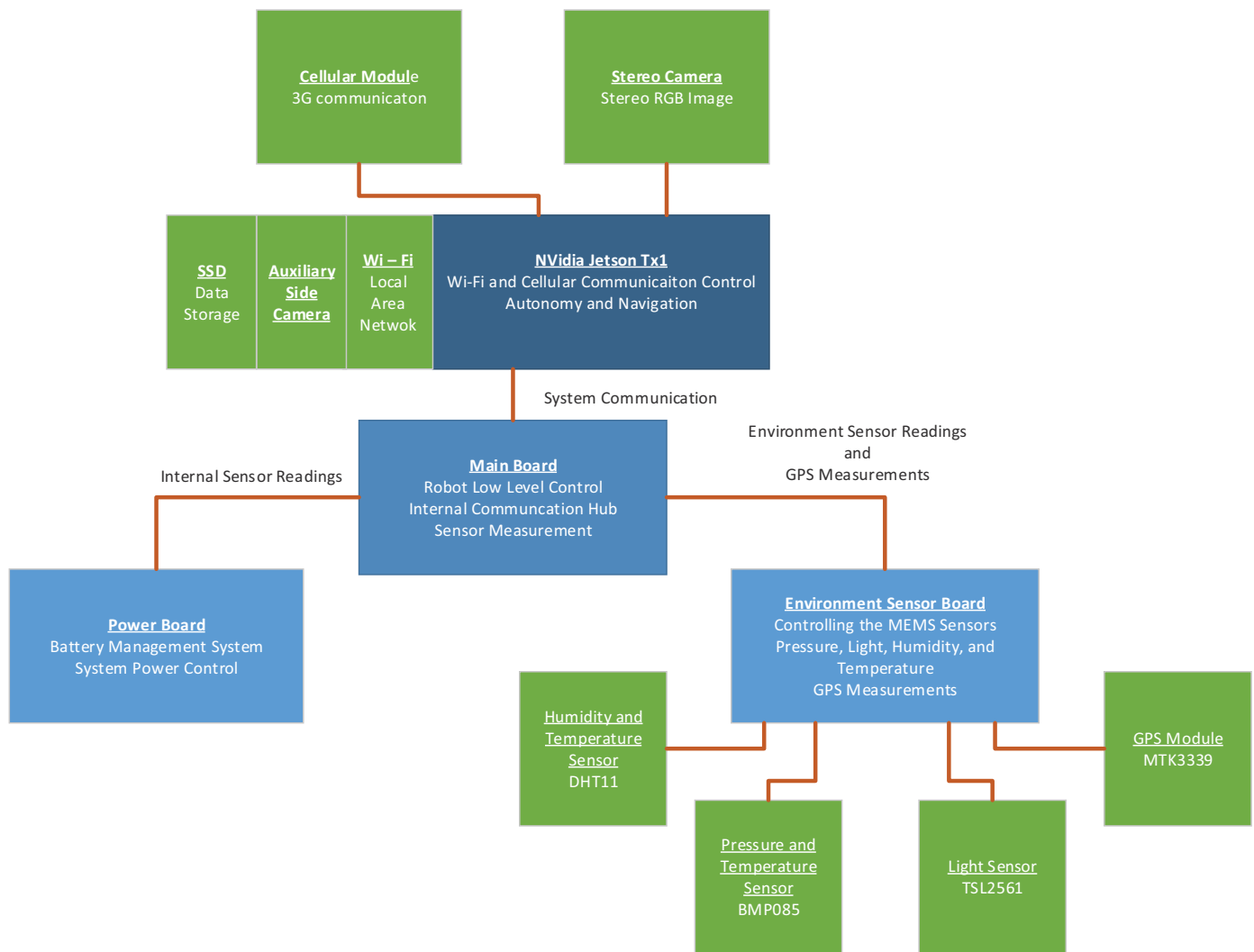


Fig. 4. System structure of the robot.

Beside these six elements, main board and NVIDIA Jetson Tx1 board are the main elements that controls the system. Tasks of these systems are given briefly at the Fig. 4.

B. Software

Software tasks with related to this work are sensor reading and communication. Other tasks like motor driving or autonomy are not in the scope of this work. System structure of the robot is shown at the Fig. 4. Brief information about all parts on the figure are given below. Green parts are not programmed; they are controlled by the linked element.

- **Power Board:** Power board works as battery management system. It monitors the battery, according to status board drives the whole system. In the meantime, it communicates with main board.

- **Environment Sensor Board:** On this boards, MEMS pressure, temperature, humidity, and light sensors are installed with Arduino microcontroller. MCU controls these sensors and sends the acquired data to main board. Additionally, MCU receives GPS NMEA packets from the GPS module and parses the packet to grab global position and time.

- **Main Board:** Main board controls the robot, measures sensors, works as communication hub. MCU, installed on the board, collects all the data that is generated by the other elements of the system, linked parts at the Figure 3.

- **NVIDIA Jetson Tx1:** Main task of this board is autonomy. With this work, in addition to autonomy high level communication task is also added. Board obtains the image from the stereo camera and the auxiliary side camera. Board selects the available network between the Wi-Fi and the cellular, and communicates with server using the WebSocket protocol or HTTP. The communication between server is explained at the system integration part.

IV. SERVER ARCHITECTURE

Fig. 5 illustrates the server architecture. As usual server consists of front-end and back-end layers. Front-end layer includes webpages; home, data, and about. Front-end layer body programmed by using HTML, CSS, and JavaScript. To speed-up development Bootstrap library is used. To visualize the data Chart.js is employed.

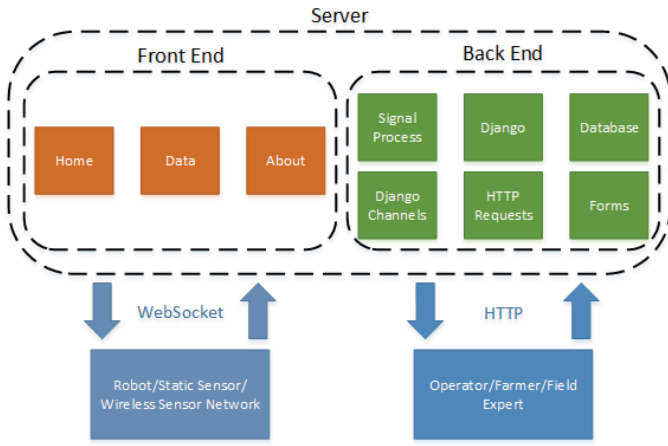


Fig. 5. Server architecture

Back-end is built-up on the Python framework Django. Django controls the database, handles HTTP requests, displays the webpages, serves forms to get data directly from the operator, enables the socket communication with WebSocket protocol of the Django-Channels module, and processes the data with additional worker.

This server architecture is created in mind that; edge processing can be applied. Edge processing is one of the aims of this server. In the literature, almost all of the processing methods are done on the python programming. Open source software can be directly applied to the server.

At the Fig. 6, screenshot from the website is shown. Here the temperature readings are shown. Since the environment sensor board has onboard GPS, location of the robot is given simultaneously. This feature shows the advantage of single mobile sensor board, instead of multiple nodes spread in field.

The visualization is important subject when the big data emerges. Here data is shown in one-line format with location. The data can be shown as marker points in map or in completely map format. The visualization is another important point that should be further improved by software engineers.

V. INTEGRATION

Server and robot are integrated through standard communication protocols. As shown at the Figure 4, first protocol is WebSocket and the second one is HTTP. In further improvements HTTPS format can be applied to make communication more secure.

HTTP request is used when the agent, robot or operator, sends the data through web browser. This protocol is used to enter manual data to the server. For the

Autonomous agents mostly use low level WebSocket protocol. Since WebSocket protocol is two-way, so that server can send the data to the robot. Besides the WebSocket protocol is asynchronous, Django-Channels module is added to the server to accomplish asynchronous data transfer.

Acquired data has four different types, that are environmental, navigational, robot status, and the image data packages from the field. Environmental data package contains

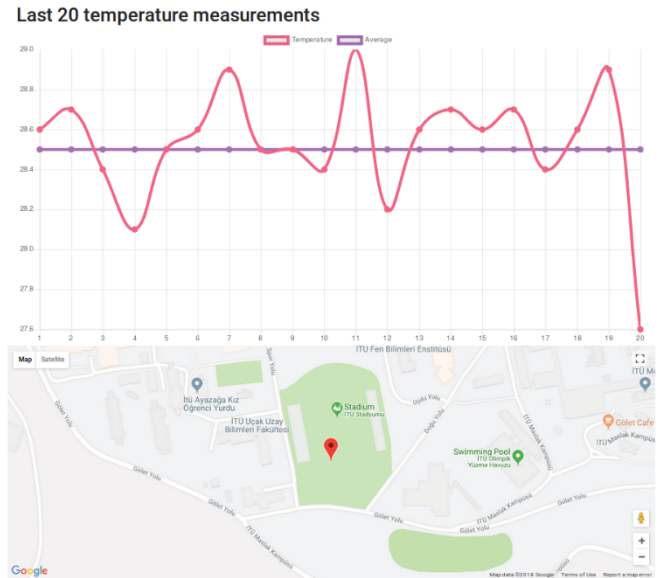


Fig. 6. Screenshot from the website.

temperature, humidity, pressure, and light measurement readings from the robot. These values are obtained by using MEMS sensors on the robot. These sensors are located openly on the robot, so that they can work easily and without interference from the robot system. Navigational data package mainly consists of GPS and the odometry data that robot generates. Third one is the robot status data package. This data package includes vital measurements from the internal sensors of the robot such as battery voltage and temperature levels. Fourth package is the image data that captured using stereo navigation camera or the auxiliary camera on the robot. First three of the explained data packages are transmitted periodically from the robot, but image data is only send if the web application asks for it. This means if the operator on the server side requires the field image, it is uploaded to the server. Package size is limited for this project needs. Any other sensor, that can be work on the agent, measurement can be send to the server. All these, packets are send inside the JSON packet as JSON array.

At the Fig. 7, whole system with WebSocket communication is shown. In this system, main board collects the data from power board, environment sensor board, and main board internal measurements. These collected data sent to Jetson Tx1 board through serial communication. Jetson Tx1 assembles the data packet in JSON format. Every JSON packet includes unique packet ID, date, packet type, and auxiliary communication byte. In addition, the sensor readings explained in the previous paragraph are added to packet in JSON array format.

The Jetson Tx1 board firstly handshakes with server. This action creates a two-way communication channel. On this channel Jetson board sends the created JSON packet to the server. When the packet reached to the server, server stores this data by creating unique identifier using date and id variables in JSON packet. For this system, after the packet sent the Jetson board closes the channel.

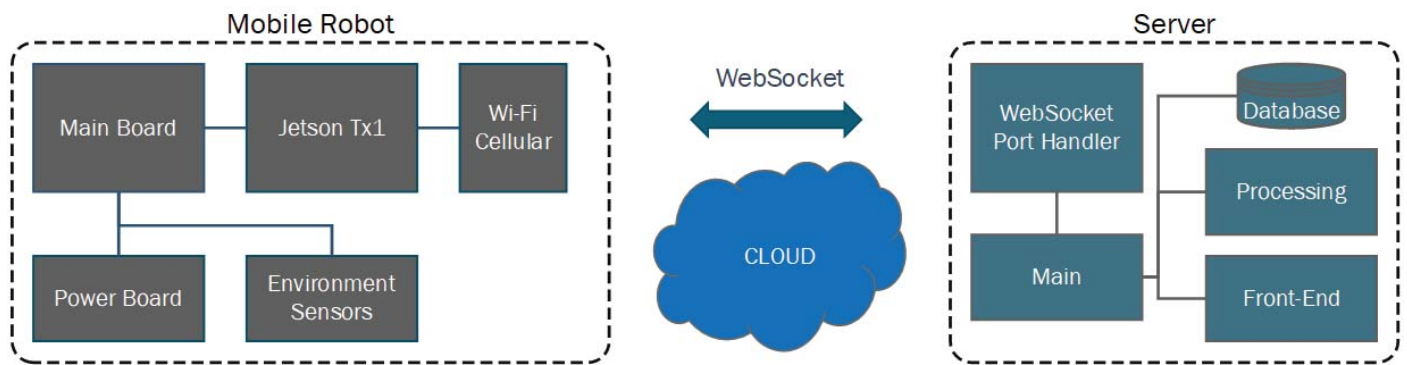


Fig. 7. System integration.

VI. CONCLUSION

This interdisciplinary work, gives know-how on intersection of robotics, internet of things, and precision agriculture. In this work, the advantage of mobile sensor platforms is explained. But the hybrid system containing mobile and static sensors is also sensible. For example, weather station as static sensor network and robot with field imaging.

The work, that is started in 2015 [13], is completed here, further advancements are classified in two sections. Research or progressive development related studies and engineering efforts.

Research will continue on this platform in different areas. Such as, developing autonomous system that can maneuver in the fields, adding precision agriculture capabilities (spraying, collecting, inspecting, ...), and mapping the field using expensive and delicate sensors such as spectrometer. Engineering efforts will be making this system a professional product or system. With software engineering can make the system robust, reliable, and usable. Lastly, good website design is also a key point to give information easily and to visualize data in more apprehensible way.

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